T7.2.3-Worksheet 2-Flow charts

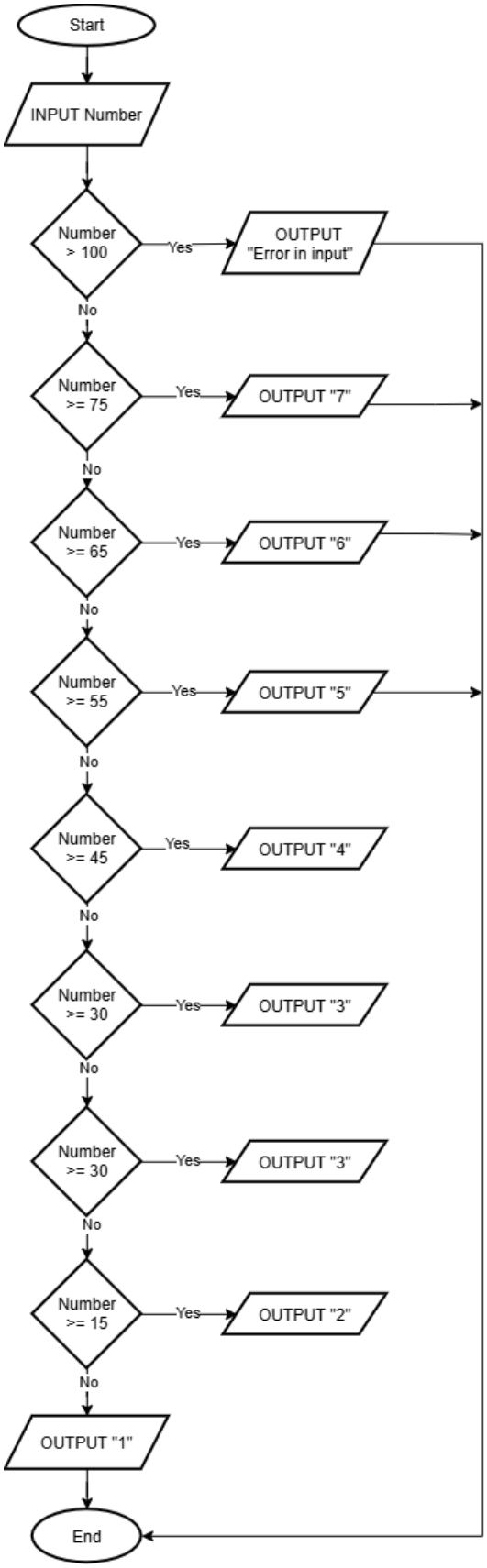
1. Selection: Draw a flow chart of an algorithm that will allow the user to:
   a. Enter a number
   b. If the number is greater than 100, display an error message.
   c. If the number is greater or equal to 75, display “7”, otherwise…
   d. If the number is greater or equal to 65, display "6”, otherwise…
   e. If the number is greater or equal to 55, display "5”, otherwise…
   f. If the number is greater or equal to 45, display "4”, otherwise…
   g. If the number is greater or equal to 30, display "3”, otherwise…
   h. If the number is greater or equal to 15, display "2”,
   i. Otherwise, display “1”

2. Counting: Draw a flow chart to check the attendance of a class:
   a. Initialise the *Absent* variable to 0
   b. Enter the number of students in a class
   c. Ask whether a student is present or absent
   d. Count the number of students who are absent (increment *Absent* by 1)
   e. Calculate the number of students who are present, based on the class size and the number of absent students
   f. Display the number of absent and present students

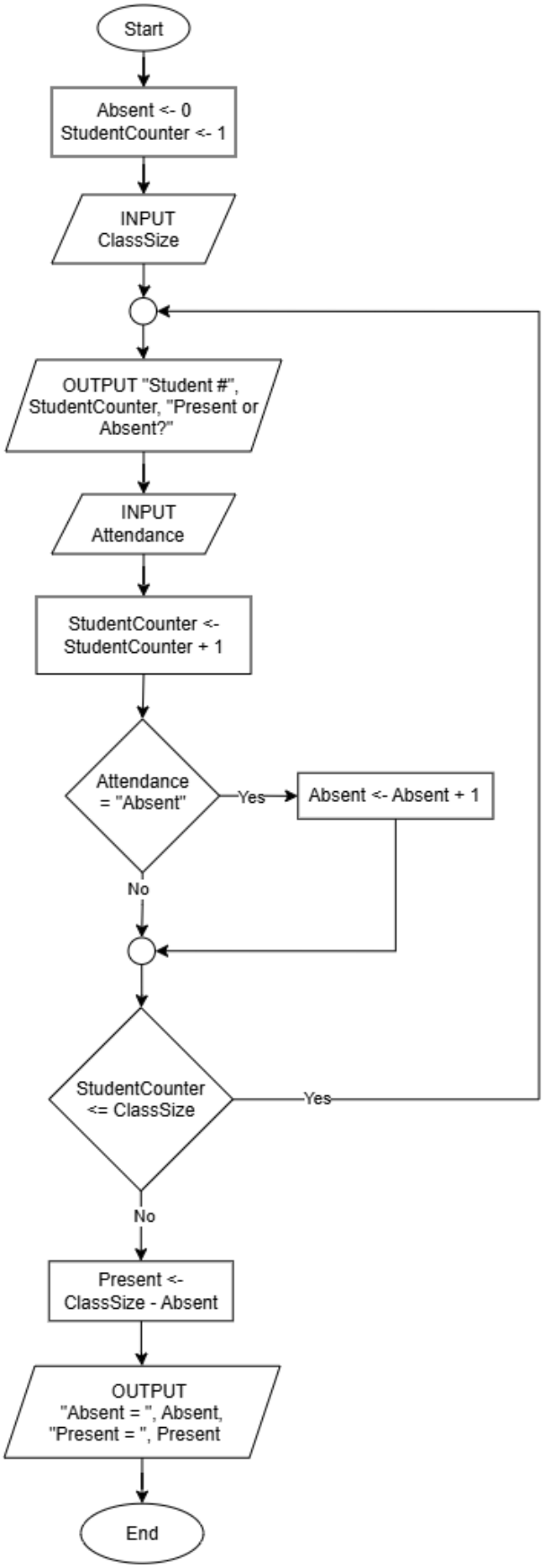
3. Draw a flow chart to calculate the sum of the digits of a number (in denary):
   a. Initialise the *Sum* variable to 0
   b. Enter a number
   c. Add the modulo (remainder) of the number and 10 to *Sum*
   d. Divide the number by 10 (integer division)
   e. Repeat steps c and d until the result of dividing the number by 10 is zero (0).
   f. Finally, display the sum of the digits of the input number.

Answers

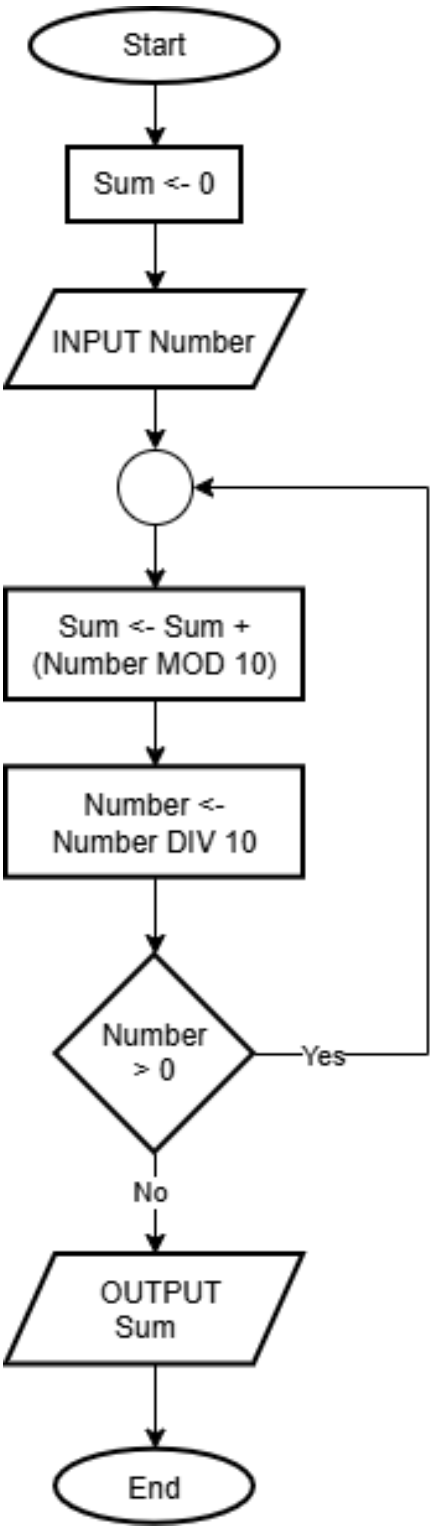
1.


Start
INPUT Number
Number > 100
Yes
OUTPUT "Error in input"
No
Number >= 75
Yes
OUTPUT "7"
No
Number >= 65
Yes
OUTPUT "6"
No
Number >= 55
Yes
OUTPUT "5"
No
Number >= 45
Yes
OUTPUT "4"
No
Number >= 30
Yes
OUTPUT "3"
No
Number >= 30
Yes
OUTPUT "3"
No
Number >= 15
Yes
OUTPUT "2"
No
OUTPUT "1"
End

2.


Start
Absent <- 0
StudentCounter <- 1
INPUT
ClassSize
OUTPUT "Student #",
StudentCounter, "Present or
Absent?"
INPUT
Attendance
StudentCounter <-
StudentCounter + 1
Attendance
= "Absent"
Yes
Absent <- Absent + 1
No
StudentCounter
<= ClassSize
Yes
No
Present <-
ClassSize - Absent
OUTPUT
"Absent = ", Absent,
"Present = ", Present
End

3.


Start
Sum <- 0
INPUT Number
Sum <- Sum +
(Number MOD 10)
Number <-
Number DIV 10
Number
> 0
Yes
No
OUTPUT
Sum
End